

Tongji Calculus, 7th Edition · Chapter 1

Functions and Limits

Exercise Workbook

100 classic-method adaptations and high-quality synthesis problems

How to Use This Workbook

The material progresses from basic to challenging. IDs Q001–Q100 match across both languages and both document types. Every formula comes from explicit LaTeX source and is compiled by the XeTeX mathematics engine.

Open-text method adaptations

20 / 100

Classic-method variants

37 / 100

Original synthesis / diagnosis

43 / 100

Source-lineage policy

Problems are redesigned around classic methods found in open textbooks, public courses, and standard calculus teaching. Copyrighted books are used only for scope and method alignment; their wording and solutions are not copied. See SOURCES.md.

1. On the first attempt, use only the exercise workbook and show complete reasoning.
2. During correction, record the cause of each error instead of copying the final answer.
3. Redo errors after 48 hours and interleave topics one week later.

Open-text method adaptation

A. For every $x \in X$, define $f(x) = x^2$

B. Define $f(-1) = 1$ and $f(1) = 1$, but do not define $f(0)$

C. Define $f(-1) = 1$ and $f(0) = 0$, while assigning both $f(1) = 0$ and $f(1) = 1$

D. For every $x \in X$, define $f(x) = x + 1$

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- Basic
- Fill in the blank
- Foundation
- Open-text method adaptation

Q002 · Common Domain of a Radical and a Logarithm

The natural domain of $f(x) = \sqrt{2 - x} + \ln(x + 1)$ is ____.

Workspace · add pages as needed

Classic-method variant

True or false: If $(f \circ g)(x_0)$ is defined, then $g(x_0)$ must belong to the domain of f .

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Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Classic-method variant

Q004 · Recognize the Rigorous Definition of a Sequence Limit

Which ε - N statement correctly expresses that the sequence $\{a_n\}$ converges to A ?

- A. For every $\varepsilon > 0$, there exists a positive integer N such that $|a_n - A| < \varepsilon$ whenever $n > N$
- B. There exists a positive integer N such that, for every $\varepsilon > 0$, $|a_n - A| < \varepsilon$ whenever $n > N$
- C. For every $\varepsilon > 0$ and every positive integer n , $|a_n - A| < \varepsilon$
- D. There exists $\varepsilon > 0$ such that, for every positive integer N , one can find $n > N$ with $|a_n - A| < \varepsilon$

Workspace · add pages as needed

- Basic
- Fill in the blank
- Foundation
- Classic-method variant

Q005 · Choose N for a Prescribed Accuracy

For $a_n = \frac{(-1)^n}{n}$, the limit is _____. When $\varepsilon = 0.01$ and the definition uses $n > N$, one may take $N =$ _____.

Workspace · add pages as needed

Classic-method variant

True or false: Every bounded sequence converges.

Workspace · add pages as needed

Open-text method adaptation

Define $f(1) = 7$, and for $x \neq 1$ let $f(x) = \frac{x^2-1}{x-1}$. What is $\lim_{x \rightarrow 1} f(x)$?

- A. 0
- B. 1
- C. 2
- D. 7

Workspace · add pages as needed

Classic-method variant

True or false: If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist and are finite, then $\lim_{x \rightarrow a} f(x)$ must exist.

True or false: If $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist and are finite, then $\lim_{x \rightarrow a} f(x)$ must exist.

- Basic
- Single choice
- Foundation
- Classic-method variant

Q009 · Recognizing an Infinitesimal

As $x \rightarrow 0$, which of the following functions is an infinitesimal?

- A. $2 + x$
- B. $\frac{x^2}{1+|x|}$
- C. $\frac{1}{x^2}$
- D. $1 + \sin x$

Workspace · add pages as needed

Classic-method variant

True or false, with justification: An infinitesimal is a nonzero number whose absolute value is very small.

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Classic-method variant

Q011 · When Can Limits Be Divided Directly?

Which conclusion follows directly from the quotient limit law?

- A. $f(x) \rightarrow 2$ and $g(x) \rightarrow 3$, so $\frac{f(x)}{g(x)} \rightarrow \frac{2}{3}$
- B. $f(x) \rightarrow 2$ and $g(x) \rightarrow 0$, so $\frac{f(x)}{g(x)} \rightarrow 0$
- C. $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$, so $\frac{f(x)}{g(x)} \rightarrow 1$
- D. Neither $f(x)$ nor $g(x)$ has a limit, so $\frac{f(x)}{g(x)}$ must have no limit

Workspace · add pages as needed

Classic-method variant

Evaluate $\lim_{x \rightarrow 2}(3x^3 - 2x^2 + x - 5)$.

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Open-text method adaptation

Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$ without derivatives.

Workspace · add pages as needed

Basic

Single choice

Foundation

Open-text method adaptation

Q014 · Recognizing the Complete Squeeze-Theorem Hypotheses

Which set of conditions is sufficient to conclude $f(x) \rightarrow L$ by the squeeze theorem?

- A. $g(x) \leq f(x) \leq h(x)$, with $g(x) \rightarrow L$ and $h(x) \rightarrow L$
- B. $g(x) \leq f(x)$, with $g(x) \rightarrow L$
- C. $f(x) \leq h(x)$, with $h(x) \rightarrow L$
- D. $g(x) \leq f(x) \leq h(x)$, but $g(x) \rightarrow L_1$ and $h(x) \rightarrow L_2$ with $L_1 < L_2$

Workspace · add pages as needed

Open-text method adaptation

Evaluate $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x}\right)$.

17 / 102.

Classic-method variant

$$\lim_{x \rightarrow 0} \frac{\sin(5x)}{3x} = \underline{\hspace{2cm}}.$$

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- Basic
- Single choice
- Foundation
- Classic-method variant

Q017 · Recognizing Equivalent Infinitesimals

As $x \rightarrow 0$, which equivalence of infinitesimals is correct?

- A. $\ln(1 + x) \sim x$
- B. $1 - \cos x \sim x$
- C. $\sin x \sim x^2$
- D. $e^x - 1 \sim 1$

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Open-text method adaptation

Q018 · The Essential Condition for Continuity

Suppose f is defined in a neighborhood of a , including at a . Which statement is equivalent to saying that f is continuous at a ?

- A. $\lim_{x \rightarrow a} f(x) = f(a)$
- B. $f(a)$ exists
- C. The left-hand limit $\lim_{x \rightarrow a^-} f(x)$ exists
- D. f is bounded in some neighborhood of a

Workspace · add pages as needed

- Basic
- Fill in the blank
- Foundation
- Open-text method adaptation

Q019 · Defining a Missing Value to Obtain Continuity

Let $f(x) = \frac{x^2-4}{x-2}$ for $x \neq 2$, and define $f(2) = c$. To make f continuous at $x = 2$, one must take $c = \underline{\hspace{1cm}}$.

Workspace · add pages as needed

- Basic
- Single choice
- Foundation
- Classic-method variant

Q020 · Finding an Interval of Continuity from the Domain

Which set is both the domain and the interval of continuity of $F(x) = \frac{\ln(4-x^2)}{\sqrt{x+1}}$?

- A. $(-2, 2)$
- B. $(-1, 2)$
- C. $[-1, 2)$
- D. $(-1, +\infty)$

Workspace · add pages as needed

- Basic
- Multiple choice
- Foundation
- Classic-method variant

Q021 · Algebra Rules for Continuous Functions

Which statements about continuity are correct?

- A. Every polynomial is continuous on $\mathbb{R}$
- B. A rational function is continuous at every point where its denominator is nonzero
- C. A composition of continuous functions is continuous wherever the composition is meaningful
- D. The quotient of two continuous functions must remain continuous at a point where the denominator function is zero

Workspace · add pages as needed

Open-text method adaptation

True or false: If f is continuous on an open interval (a, b) , then it must attain both a maximum and a minimum in (a, b) .

True or false: If f is continuous on an open interval (a, b) , then it must attain both a maximum and a minimum in (a, b) .

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Open-text method adaptation

Q023 · A Composite Function and Its Domain

Let $f(x) = \sqrt{x + 2}$ and $g(u) = \frac{1}{u-1}$. Find the formula and natural domain of $(g \circ f)(x)$.

Workspace · add pages as needed

- Standard
- Multiple choice
- Methods
- Classic-method variant

Q024 · Analyze Global Properties of a Function

Let $h(x) = \frac{|x|}{1+x^2}$ on $\mathbb{R}$. Which statements are correct?

- A. h is even
- B. For every $x \in \mathbb{R}, 0 \leq h(x) \leq \frac{1}{2}$
- C. h is periodic
- D. h is strictly increasing on $[0, +\infty)$

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Open-text method adaptation

Q025 · An Inverse Function with an Interval Restriction

The domain of $f(x) = 2x - 3$ is $[1, 4]$. Find the range of f , the formula for its inverse, and the domain and range of the inverse.

Workspace · add pages as needed

- Standard
- Single choice
- Methods
- Classic-method variant

Q026 · Find the Least Positive Period of a Sum

What is the least positive period of $p(x) = \sin(2x) + \cos(4x)$?

- A. $\frac{\pi}{4}$
- B. $\frac{\pi}{2}$
- C. π
- D. 2π

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q027 · Limit of a Rational Sequence with Equal Degrees

Evaluate $\lim_{n \rightarrow \infty} \frac{5n^2 - 3n + 1}{2n^2 + n - 4}$.

Workspace · add pages as needed

- Standard
- Proof
- Methods
- Classic-method variant

Q028 · Prove Convergence of a Rational Sequence from the Definition

Use the ε - N definition to prove $\lim_{n \rightarrow \infty} \frac{3n-2}{n+1} = 3$.

Workspace · add pages as needed

- Standard
- Fill in the blank
- Methods
- Classic-method variant

Q029 · Determine a Parameter from One-Sided Limits

Let $f(x) = \begin{cases} 2x + a, & x < 1, \\ x^2 + 1, & x > 1. \end{cases}$ For $\lim_{x \rightarrow 1} f(x)$ to exist, a must equal ____, and the limit is then ____.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q030 · Use One-Sided Limits to Show Nonexistence

Evaluate $\lim_{x \rightarrow 0} \frac{|x|}{x}$. If it does not exist, explain why.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q031 · Handle the Sign of a Radical as x Approaches Negative Infinity

Evaluate $\lim_{x \rightarrow -\infty} \frac{2x-1}{\sqrt{x^2+1}}$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Open-text method adaptation

Q032 · Cancel the Common Factor Causing Zero Division

Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.

Workspace · add pages as needed

Classic-method variant

As $x \rightarrow 0$, write $f(x) = \frac{3+2x}{1+x}$ in the form $f(x) = A + \alpha(x)$, where $\alpha(x)$ is infinitesimal. Find A and $\alpha(x)$.

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Classic-method variant

Find $\lim_{x \rightarrow 0} \frac{1}{x^2}$, and explain why the reciprocal relationship between $\frac{1}{x^2}$ and x^2 is valid as $x \rightarrow 0$.

Open-text method adaptation

Evaluate $\lim_{x \rightarrow 1} \frac{\sqrt{x+3}-2}{x-1}$ without derivatives.

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Classic-method variant

Evaluate $\lim_{x \rightarrow 8} \frac{\sqrt[3]{x}-2}{x-8}$ without derivatives.

- Standard
- Fill in the blank
- Methods
- Classic-method variant

Q037 · A Rational Expression from Known Limits

If $f(x) \rightarrow 2$ and $g(x) \rightarrow -1$ as $x \rightarrow a$, then $\lim_{x \rightarrow a} \frac{3f(x)-g(x)}{f(x)-g(x)} = \underline{\hspace{1cm}}$.

Workspace · add pages as needed

- Standard
- True/false with justification
- Methods
- Classic-method variant

Q038 · A Quotient of Two Infinitesimals

True or false, with justification: If $\alpha(x) \rightarrow 0$ and $\beta(x) \rightarrow 0$, then $\frac{\alpha(x)}{\beta(x)} \rightarrow 1$.

Workspace · add pages as needed

Classic-method variant

For $L = \lim_{x \rightarrow -\infty} \frac{3x^2 - x + 1}{2x^2 + 5}$, which statements are correct? Select all that apply.

D. Since $x \rightarrow -\infty$, $x^2 \rightarrow -\infty$

Open-text method adaptation

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(2x) \sin(3x)}{x^2}$.

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Classic-method variant

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x^2}$ without Taylor expansions.

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Classic-method variant

Evaluate $\lim_{x \rightarrow 0} (1 + 2x)^{\frac{3}{x}}$.

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Classic-method variant

Evaluate $\lim_{n \rightarrow \infty} (1 - \frac{3}{n})^n$.

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- Standard
- Multiple choice
- Methods
- Classic-method variant

Q044 · Recognizing Common Indeterminate Forms

Which limit structures are indeterminate forms? Select all that apply.

- A. $\frac{0}{0}$
- B. $\frac{\infty}{\infty}$
- C. 1^∞
- D. 0^1

Workspace · add pages as needed

- Standard
- Fill in the blank
- Methods
- Classic-method variant

Q045 · Comparing Orders of Power Infinitesimals

As $x \rightarrow 0$, $|x|^3$ is a ____-order infinitesimal relative to x^2 because $\frac{|x|^3}{x^2} \rightarrow$ ____.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Classic-method variant

Q046 · Equivalent Substitution in a Product-Quotient Structure

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x) \cdot \ln(1+2x)}{x(e^x-1)}$.

Workspace · add pages as needed

Classic-method variant

Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos(2x)}{x \sin(3x)}$.

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Open-text method adaptation

Workspace · add pages as needed

- Standard
- Multiple choice
- Methods
- Classic-method variant

Q050 · Distinguishing Types of Discontinuity

Which statements about a discontinuity at $x = a$ are correct?

- A. If both one-sided limits exist, are equal and finite, but their common value differs from $f(a)$ (or $f(a)$ is undefined), then a is a removable discontinuity
- B. If both one-sided limits are finite but unequal, then a is a jump discontinuity
- C. If at least one one-sided limit is infinite, then a is an infinite discontinuity and is of the second kind
- D. If the limit fails to exist because of oscillation, then a is a discontinuity of the first kind

Workspace · add pages as needed

- Standard
- Proof
- Methods
- Classic-method variant

Q051 · Proving Continuity of a Quadratic from the Definition

Without citing the theorem that polynomials are continuous, use the ε - δ definition to prove that $f(x) = x^2 + 1$ is continuous at every $a \in \mathbb{R}$.

Workspace · add pages as needed

Open-text method adaptation

Use continuity to evaluate $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\ln(2 + \cos x)}{\sqrt{1 + \sin x}}$, and explain why direct substitution is valid.

Use continuity to evaluate $\lim_{x \rightarrow \frac{\pi}{6}} \frac{\ln(2 + \cos x)}{\sqrt{1 + \sin x}}$, and explain why direct substitution is valid.

- Standard
- Proof
- Methods
- Classic-method variant

Q053 · The Square Root of a Positive Continuous Function

Suppose f is continuous at a and $f(a) > 0$. Prove that in some neighborhood of a , $f(x) > \frac{f(a)}{2} > 0$, and then prove that $g(x) = \sqrt{f(x)}$ is continuous at a .

Workspace · add pages as needed

Standard

Fill in the blank

Methods

Open-text method adaptation

Q054 · A Value Between the Endpoint Values

Suppose f is continuous on $[0, 2]$, with $f(0) = -3$ and $f(2) = 5$. By the intermediate value theorem, there is at least one $\xi \in \rule{1cm}{0.4pt}$ such that $f(\xi) = 1$.

Workspace · add pages as needed

- Standard
- Calculation
- Methods
- Open-text method adaptation

Q055 · Locating and Proving Uniqueness of a Cubic Root

Without using derivatives, prove that $x^3 + x - 1 = 0$ has exactly one real root in $(\frac{1}{2}, \frac{3}{4})$.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q056 · Parity of a Composite Function

Let D be symmetric about the origin. Suppose $g : D \rightarrow E$ is even, $f : E \rightarrow R$ is arbitrary, and $h = f \circ g$ is defined on D . Prove that h is even on D .

Workspace · add pages as needed

- Advanced
- Parameter / synthesis / application
- Synthesis
- Original synthesis / diagnosis

Q057 · A Piecewise Parking-Fee Model

A parking lot charges by continuous parking time t in hours. For $0 < t \leq 2$, the fee is 6 yuan. For $2 < t \leq 5$, it charges 6 yuan plus 3 yuan per hour for the portion beyond 2 hours. Write the piecewise formula $C(t)$, find its domain and range, and describe its monotonicity and boundedness.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q058 · Rationalize a Difference of Radicals

Evaluate $\lim_{n \rightarrow \infty} [\sqrt{n^2 + 3n} - n]$.

Workspace · add pages as needed

Advanced

Proof

Synthesis

Original synthesis / diagnosis

Q059 · Eventual Positivity from a Positive Limit

Suppose $a_n \rightarrow a$ and $a > 0$. Prove that there exists a positive integer N such that $a_n > \frac{a}{2}$ whenever $n > N$.

Workspace · add pages as needed

- Advanced
- Multiple choice
- Synthesis
- Original synthesis / diagnosis

Q060 · Necessary Consequences of a Known Limit

Suppose $a_n \rightarrow 2$. Which conclusions must be true?

- A. $\{a_n\}$ is bounded
- B. There exists N such that $a_n > 0$ whenever $n > N$
- C. There exists N such that the tail a_n is strictly increasing
- D. Every subsequence of $\{a_n\}$ converges to 2

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q061 · An Epsilon-Delta Proof for a Quadratic Limit

Use the epsilon-delta definition to prove $\lim_{x \rightarrow 2}(x^2 + 1) = 5$.

Workspace · add pages as needed

Original synthesis / diagnosis

Fill in the blanks: $\lim_{x \rightarrow 2^-} \frac{1}{x-2} = \underline{\hspace{1cm}}$, $\lim_{x \rightarrow 2^+} \frac{1}{x-2} = \underline{\hspace{1cm}}$, and therefore $\lim_{x \rightarrow 2} \frac{1}{x-2} \underline{\hspace{1cm}}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q063 · Limit of a Difference of Two Radicals

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q064 · Parameter Classification with an Oscillatory Factor

Let $f_p(x) = |x|^p \sin(\frac{1}{x})$ for $x \neq 0$. For $p > 0$, $p = 0$, and $p < 0$, decide whether f_p is infinitesimal, bounded with no limit, or unbounded without tending to $+\infty$ or $-\infty$ as $x \rightarrow 0$.

Workspace · add pages as needed

- Advanced
- True/false with justification
- Synthesis
- Original synthesis / diagnosis

Q065 · Does Local Unboundedness Imply an Infinite Limit?

True or false, with justification: If f is unbounded in every deleted neighborhood of $x = 0$, then $\lim_{x \rightarrow 0} |f(x)| = +\infty$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q066 · Double Rationalization of Nested Radicals

Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+\sqrt{1+x}}-\sqrt{2}}{x}$ without derivatives or Taylor expansions.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q067 · Determining Parameters from a Finite Limit

Find real numbers a,b such that $\lim_{x \rightarrow 1} \frac{x^2+ax+b}{x-1} = 3$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q068 · Using a Known Ratio of Deviations

As $x \rightarrow a$, suppose $f(x) \rightarrow 1$, $g(x) \rightarrow 2$, and $\frac{f(x)-1}{g(x)-2} \rightarrow 3$. Find $\lim_{x \rightarrow a} \frac{f(x)^2-1}{g(x)^2-4}$.

Workspace · add pages as needed

Advanced

Proof

Synthesis

Original synthesis / diagnosis

Q069 · Proving the Product Limit Theorem from Definitions

Suppose $f(x) \rightarrow A$ and $g(x) \rightarrow B$ as $x \rightarrow a$. Using only the limit definition and local boundedness of convergent functions, prove that $f(x)g(x) \rightarrow AB$.

Workspace · add pages as needed

Original synthesis / diagnosis

Evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x)}{\sin(5x)}$.

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- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q071 · The Second Fundamental Limit with a Fractional Base

Evaluate $\lim_{x \rightarrow 0} \left[\frac{1+x}{1-x} \right]^{\frac{1}{x}}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q072 · Determining Parameters from an Exponential Limit

Let a, b be positive real numbers. Given $\lim_{x \rightarrow 0} (1 + ax)^{\frac{b}{x}} = e^6$ and $a + b = 5$, find all ordered pairs (a, b) .

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q073 · Squeezing the Floor-Function Error

Use the squeeze theorem to prove $\lim_{x \rightarrow +\infty} \frac{\lfloor x \rfloor}{x} = 1$.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q074 · A Monotone-Bounded Proof for a Linear Recurrence

Let $a_1 > 2$ and $a_{n+1} = \frac{a_n+2}{3}$. Prove that $\{a_n\}$ converges and find its limit, without first deriving a closed form.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q075 · A Mixed Exponential-Logarithmic-Trigonometric Limit

Evaluate $\lim_{x \rightarrow 0} \frac{(e^{2x}-1) \tan(3x)}{x \ln(1+4x)}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q076 · Determining Parameters from an Equivalence

Let $a, b > 0$. If $e^{ax} - 1 \sim b \sin(2x)$ as $x \rightarrow 0$ and $a + b = 3$, find a and b.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q077 · Checking a First-Order Remainder in a Difference

Without Taylor expansions, evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{1+x}-1-\frac{x}{2}}{x}$.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Open-text method adaptation

Q078 · Three Types of Discontinuity in One Function

Let $f(x) = -1$ for $x < 0$; $f(x) = \frac{x^2-1}{x-1}$ for $0 \leq x < 2$ with $x \neq 1$; and $f(x) = \frac{1}{x-2}$ for $x > 2$. Find and classify every discontinuity. If one is removable, give the value for a continuous extension.

Workspace · add pages as needed

- Advanced
- Error diagnosis
- Synthesis
- Original synthesis / diagnosis

Q079 · Must the Sum of Two Discontinuous Functions Be Discontinuous?

A student claims: “If both f and g are discontinuous at $x = 0$, then $f + g$ must be discontinuous at 0, because only sums of continuous functions are continuous.” Identify the logical error and construct and verify a counterexample.

Workspace · add pages as needed

- Advanced
- Proof
- Synthesis
- Original synthesis / diagnosis

Q080 · Finding the Unique Junction Value by One-Sided Continuity

Fix a real number a . Define $f(x) = \frac{\sin(x-a)}{x-a}$ for $x < a$, $f(a) = c$, and $f(x) = \frac{\ln(1+x-a)}{x-a}$ for $x > a$. First prove that a piecewise function is continuous at a exactly when both one-sided limits equal $f(a)$, and then find the unique c making this function continuous.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q081 · Finding Closed-Interval Extrema without Derivatives

Without using derivatives, find the maximum and minimum of $f(x) = x^4 - 2x^2 + 3$ on $[-2, 2]$, together with all points where they are attained.

Workspace · add pages as needed

- Advanced
- Calculation
- Synthesis
- Original synthesis / diagnosis

Q082 · Finding a Complete Range from Bounds and the Intermediate Value Property

Without using derivatives, find the range of $f(x) = x + \frac{4}{x+1}$ on $[0, 3]$, and explain why every number between the two bounds is attained.

Workspace · add pages as needed

Hard

Error diagnosis

Synthesis

Original synthesis / diagnosis

Q083 · Diagnose an Endpoint Error in a Domain

Find the domain of $F(x) = \ln(1 - \sqrt{x - 1})$. A student writes: “The radical requires $x \geq 1$. The logarithm requires $1 - \sqrt{x - 1} \geq 0$, so $x \leq 2$. Hence the domain is $[1, 2]$.” Identify the error and give the correct solution.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q084 · Inverse of a Composition of Bijections

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be bijections. Prove that $g \circ f : A \rightarrow C$ is a bijection and that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Workspace · add pages as needed

Hard

Error diagnosis

Synthesis

Original synthesis / diagnosis

Q085 · A Limit Equation Cannot Replace a Convergence Proof

Let $a_1 = 2$ and $a_{n+1} = \sqrt{3 + a_n}$. A student writes: “Assume $a_n \rightarrow L$. Then $L = \sqrt{3 + L}$, so $L = \frac{1+\sqrt{13}}{2}$. Therefore the sequence converges to this value.” Identify the logical error and complete a rigorous proof.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q086 · Prove Uniqueness of a Function Limit

Let a be an accumulation point of the domain of f . If $\lim_{x \rightarrow a} f(x) = A$ and $\lim_{x \rightarrow a} f(x) = B$, prove that $A = B$.

Workspace · add pages as needed

Hard

Proof

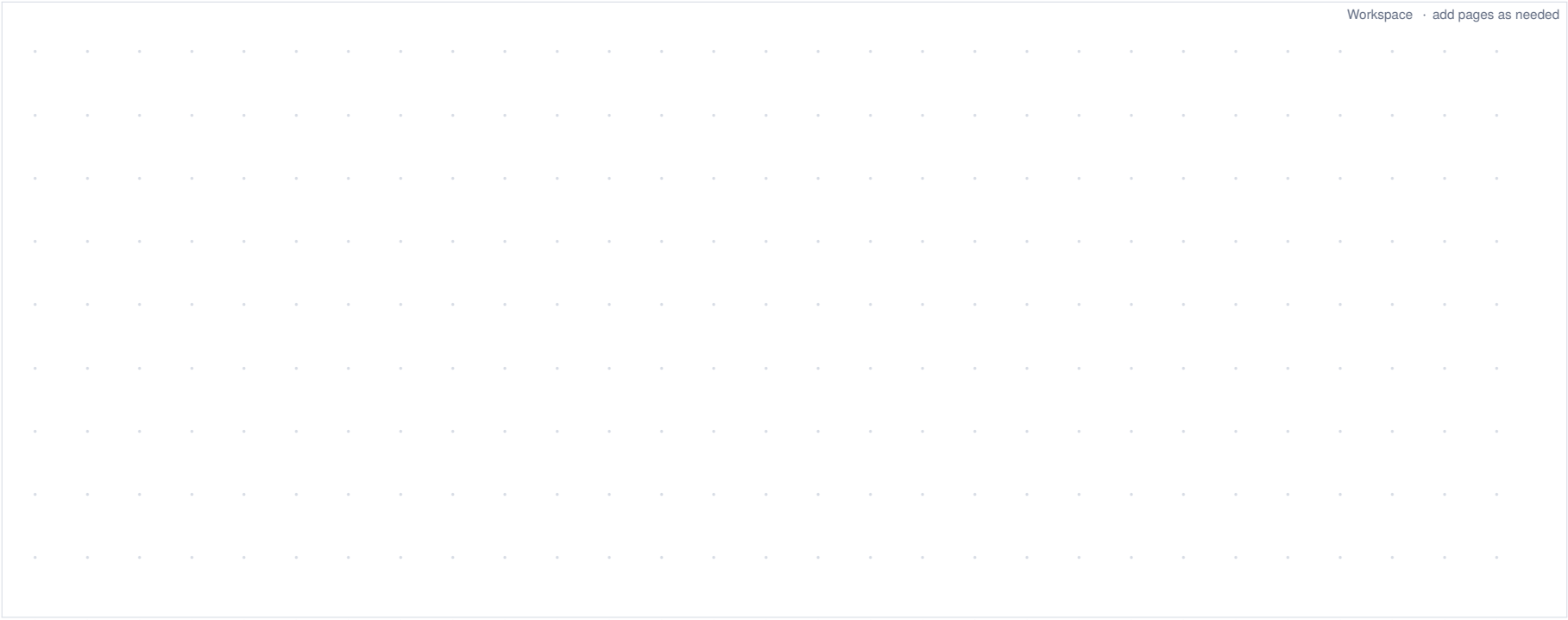
Synthesis

Original synthesis / diagnosis

Q087 · A Bounded Function Times an Infinitesimal

Suppose $\alpha(x)$ is infinitesimal as $x \rightarrow a$ and $g(x)$ is bounded in a deleted neighborhood of a . Using definitions only, prove that $\alpha(x)g(x)$ is also infinitesimal, treating separately the case where the bound may be 0.

Workspace · add pages as needed



Hard

Error diagnosis

Synthesis

Original synthesis / diagnosis

Q088 · Diagnosing Errors in $\frac{0}{0}$ and Cancellation

A student evaluates $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$ as follows: first, substitution gives $\frac{0}{0} = 0$; second, canceling $x - 1$ gives $x + 1$, so the limit is also 2; hence the answer is both 0 and 2. Identify the logical error in each step and give the complete correct solution.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Original synthesis / diagnosis

Q089 · A Variable-Exponent Limit at Infinity

Evaluate $\lim_{x \rightarrow +\infty} (1 + \frac{3}{x})^{2x - \sqrt{x}}$, explicitly identifying the 1^∞ form and the effective standardized exponent.

Workspace · add pages as needed

Hard

Proof

Synthesis

Open-text method adaptation

Q090 · A Geometric Proof of the First Fundamental Limit

Assuming radian measure, start from $\sin x < x < \tan x$ for $0 < x < \frac{\pi}{2}$ and prove $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, covering both $x \rightarrow 0^+$ and $x \rightarrow 0^-$.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q091 · Equivalence Characterized by Relative Error

Let $\alpha(x)$ and $\beta(x)$ both be infinitesimal as $x \rightarrow a$, with $\beta(x)$ nonzero in a deleted neighborhood. Prove that $\alpha(x) \sim \beta(x)$ if and only if $\alpha(x) - \beta(x) = o(\beta(x))$.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q092 · Proving Safe Replacement in a Quotient

Suppose $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ as $x \rightarrow a$, with all relevant quotients defined in a deleted neighborhood. Prove $\frac{\alpha}{\beta} \sim \frac{\alpha_1}{\beta_1}$, and explain why this supports equivalent substitution in products and quotients.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q093 · A Counterexample to Replacement inside a Sum

Construct and verify a counterexample showing that $\alpha \sim \alpha_1$ and $\beta \sim \beta_1$ need not imply $\alpha + \beta \sim \alpha_1 + \beta_1$. Require both sums to be nonzero in a deleted neighborhood.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Original synthesis / diagnosis

Q094 · Can a Power Factor Suppress Oscillation?

For a real parameter p , define $f_p(0) = 0$ and $f_p(x) = |x|^p \sin(\frac{1}{x})$ for $x \neq 0$. Discuss continuity at $x = 0$. When it is discontinuous, classify the discontinuity and decide whether the function is bounded in some neighborhood of 0.

Workspace · add pages as needed

Hard

Proof

Synthesis

Original synthesis / diagnosis

Q095 · Equal Values Half an Interval Apart

Suppose f is continuous on $[0, 1]$ and $f(0) = f(1)$. Prove that there is at least one $\xi \in [0, \frac{1}{2}]$ such that $f(\xi) = f(\xi + \frac{1}{2})$.

Workspace · add pages as needed

Hard

Parameter / synthesis / application

Synthesis

Original synthesis / diagnosis

Q096 · Optional: Comparing Uniform Continuity on an Unbounded Interval

On $[0, +\infty)$, determine whether $f(x) = \sqrt{x}$ and $g(x) = x^2$ are uniformly continuous. Prove both conclusions directly from the definition. For the function that is not uniformly continuous, give a specific ε_0 and a contradiction construction.

Workspace · add pages as needed

Challenge

Parameter / synthesis / application

Challenge

Original synthesis / diagnosis

Q097 · Characterize Parameter Convergence Using Even and Odd Subsequences

For real p and q , define $a_{2k} = p + \frac{1}{k}$ and $a_{2k-1} = q - \frac{1}{k}$ for $k = 1, 2, \dots$. Find a necessary and sufficient condition for $\{a_n\}$ to converge. In the convergent case, state the limit and prove it using the ε - N definition.

Workspace · add pages as needed

Challenge

Parameter / synthesis / application

Challenge

Original synthesis / diagnosis

Q098 · A Complete Convergence Chain for a Radical Recurrence

Let $0 < a_1 < 2$ and $a_{n+1} = \sqrt{2 + a_n}$. Prove: (1) $0 < a_n < 2$; (2) a_n is increasing; (3) it converges; and (4) find its limit. Do not guess a closed form or merely solve a limit equation.

Workspace · add pages as needed

Challenge

Parameter / synthesis / application

Challenge

Original synthesis / diagnosis

Q099 · Safe and Unsafe Zones for Equivalent Substitution

Complete the following without derivatives, L'Hôpital's rule, or Taylor expansions. (1) Evaluate $A = \lim_{x \rightarrow 0} \frac{\sin(2x)}{e^x - 1}$. (2) A student uses $\sqrt{1+x} - 1 \sim \frac{x}{2}$ to replace $\sqrt{1+x} - 1$ inside a difference and claims $B = \lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2} = 0$. Judge the method and conclusion, then find B by exact algebra. (3) State one rule describing when direct equivalent-infinitesimal substitution is valid.

Workspace · add pages as needed

Challenge

Parameter / synthesis / application

Challenge

Original synthesis / diagnosis

Q100 · Optional: The Uniform-Continuity Boundary for Oscillatory Functions

On $(0, 1]$, define $F(x) = x \sin(\frac{1}{x})$ and $G(x) = \sin(\frac{1}{x})$. Determine whether each is uniformly continuous. For F , use a continuous extension and the uniform-continuity theorem on a closed interval. For G , construct two sequences whose distance tends to 0 while their function values differ by exactly 2.

Workspace · add pages as needed